

LOW STYRENE SBR PATENT LANDSCAPE: SOLUTION-POLYMERIZED AND MODIFIED STYRENE-BUTADIENE COPOLYMERS

1. ABSTRACT

This report synthesizes three primary patents centered on styrene-butadiene rubber/copolymer materials relevant to low styrene SBR. The selected patents span two major synthesis families: solution polymerization of styrene and butadiene in hydrocarbon solvent, and post-polymerization modification of solution-made styrene-butadiene copolymers with functional modifiers. Across the set, the target material is not a block styrenic elastomer but a random or tapered styrene-butadiene copolymer designed for tire-reinforcement performance, with styrene content explicitly controlled in the low-to-moderate range. One patent discloses a low molecular weight tapered styrene-butadiene copolymer with total styrene content of 20 to 30 wt% and a random/tapered sequence distribution intended to replace process oil in rubber compounds. Another patent discloses a modified styrene-butadiene copolymer prepared by anionic solution polymerization in hexane followed by terminal modification with a silane-containing modifier, emphasizing filler affinity and processability. The third patent discloses a modified conjugated diene-based polymer prepared by polymerizing styrene and butadiene in hydrocarbon solvent with an organometallic catalyst system, then modifying the active polymer with an alkoxysilane-based modifier; it explicitly targets a styrene-butadiene random copolymer with 50 to 95 wt% diene and 5 to 50 wt% aromatic vinyl in one embodiment. Major process trends include hydrocarbon-solvent solution polymerization, organolithium or mixed catalyst initiation, use of polar modifiers/additives, controlled monomer feed and temperature, and downstream termination/coagulation/drying steps. Water is not part of the core polymerization medium in any of the three patents, but appears in workup or recovery steps in two patents. Overall, the landscape emphasizes composition control, sequence distribution, and post-polymerization functionalization rather than simply lowering styrene content alone, making these patents technically relevant to low styrene SBR synthesis and tire-compound performance optimization.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10174133B2
- Patent Title: Modified styrene-butadiene copolymer, preparation method thereof, and rubber ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in normal hexane using n-butyllithium and a polar additive, followed by terminal capping with butadiene and post-polymerization modification with a silane-containing modifier. The polymer is then terminated with ethanol and recovered by water-assisted solvent removal and roll drying. - Styrene charge: 29.7 g — Example 1 - 1,3-Butadiene charge: 78.1 g — Example 1 - Normal hexane charge: 550 g — Example 1 - Polar additive (2,2-di(2-tetrahydrofuryl)propane): 0.124 g — Example 1 - Reaction temperature: 50° C. — Example 1 - n-Butyllithium: 3 mmol — Example 1 - Terminal capping butadiene: 2.2 g — Example 1 - Modifier amount: 0.181 g — Example 1 - [DTP]/[act. Li]: 2.3 — Example 1 - [modifier]/[act. Li]: 1.0 — Example 1 - Styrene charge: 270 g — Example 2 - 1,3-Butadiene charge: 710 g — Example 2 - Normal hexane charge: 5 kg — Example 2 - Polar additive (2,2-di(2-tetrahydrofuryl)propane): 0.94 g — Example 2 - Reaction temperature: 40° C. — Example 2 - n-Butyllithium: 25.40 g (2.62 wt % in hexane, 33% activation) — Example 2 - Terminal capping butadiene: 20.0 g — Example 2 - Modifier amount: 1.98 g — Example 2 - [DTP]/[act. Li]: 1.46 — Example 2 - [modifier]/[act. Li]: 0.85 — Example 2 - Antioxidant solution: 0.3 wt % BHT in hexane — Example 1 - Antioxidant solution: 30 wt % Wingstay K in hexane — Example 2 - Medium & Water Role: Core polymerization is carried out in normal hexane solution. Water is used later only when the polymer is added to warm water heated using steam to remove solvents, followed by removal of remaining solvents and water by roll drying. - styrene content: Not disclosed in extracted evidence

Medium & Water Role - Core polymerization is carried out in normal hexane solution. Water is used later only when the polymer is added to warm water heated using steam to remove solvents, followed by removal of remaining solvents and water by roll drying. - Core reaction medium: Organic hydrocarbon solution polymerization in normal hexane

styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Produced a modified styrene-butadiene copolymer from styrene/butadiene in hexane at 50° C. with n-butyllithium and DTP, then modified it with a silane-containing agent. - Example 2: Demonstrated the same solution-polymerization/modification sequence at larger scale and 40° C., using a different silane-containing modifier. - Comparative Example: Showed that using N,N-bis(triethoxysilylpropyl)piperazine as modifier gave a comparative modified copolymer. - Experimental Example 1: Showed higher combination efficiency for Examples 1 and 2 versus the

comparative sample.

Technical Relevance This patent is directly relevant because it prepares a styrene-butadiene copolymer by solution polymerization and explicitly controls monomer charge, temperature, and post-modification. Although not a low-styrene-specific patent, it provides a route to functionalized SBR with composition and process control important for low styrene grades.

Patent 2

- Patent Number: US11591428B2
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and butadiene in a hydrocarbon solvent using an organolithium initiator and sodium dodecylbenzenesulfonate, producing a low molecular weight tapered styrene-butadiene copolymer. The polymer is recovered by coagulation in isopropanol containing BHT and drum drying, and is then used in rubber compositions as a partial or full oil replacement. - Total styrene content: 20 to 30 percent by weight — Claim 1 - Total butadiene content: 70 to 80 percent by weight — Claim 1 - Styrene sequences >3 units: greater than 30 weight percent of total styrene content — Claim 1 - Styrene sequences >10 units: less than 20 weight percent of total styrene content — Claim 1 - Number average molecular weight: 30,000 to 80,000 — Claim 1 - Total styrene content: 22 to 27 percent by weight — Claim 2 - Total butadiene content: 73 to 78 percent by weight — Claim 2 - Styrene sequences >3 units: 35 to 50 weight percent of total styrene content — Claim 3 - Styrene sequences >10 units: 8 to 18 weight percent of total styrene content — Claim 4 - Glass transition temperature: -80° C. to -70° C. — Claim 5 - Molecular weight distribution: 1.05 to 1.13 — synthesis passage - Rubber composition filler: 30 to 100 phr — synthesis passage - Process oil: up to 10 phr — synthesis passage - Low molecular weight styrene-butadiene copolymer: 1 to 15 phr — synthesis passage - Polymerization temperature: 30° C. to 80° C. — synthesis passage - Polymerization temperature: 40° C. to 70° C. — synthesis passage - Polymerization temperature: 50° C. to 60° C. — synthesis passage - Peak polymerization temperature: less than 120° C. — synthesis passage - Peak polymerization temperature: less than 110° C. — synthesis passage - Peak polymerization temperature: less than 100° C. — synthesis passage - Peak polymerization temperature: less than 90° C. — synthesis passage - Peak polymerization temperature: less than 80° C. — synthesis passage - Peak polymerization temperature: less than 70° C. — synthesis passage - Peak polymerization temperature: less than 60° C. — synthesis passage - Peak polymerization temperature: less than 50° C. — synthesis passage - Polymerization time: 30 minutes to 60 minutes to several hours — synthesis passage - Residence time: 30 minutes to 2 hours — synthesis passage - Residence

time: 30 minutes to 90 minutes — synthesis passage - Desolventization temperature: 100° C. to 170° C. — synthesis passage - Medium & Water Role: Core polymerization is solution-based in a hydrocarbon solvent. Water is not the polymerization medium; it appears later only in coagulation workup through isopropanol-containing coagulation and drum drying of the recovered polymer. - styrene content: 20 to 30 percent by weight; preferred 22 to 27 percent by weight — Claim 1, Claim 2

Medium & Water Role - Core polymerization is solution-based in a hydrocarbon solvent. Water is not the polymerization medium; it appears later only in coagulation workup through isopropanol-containing coagulation and drum drying of the recovered polymer. - Core reaction medium: Organic hydrocarbon solution polymerization in solvent

styrene content - 20 to 30 percent by weight; preferred 22 to 27 percent by weight — Claim 1, Claim 2

Relevant Experimental Evidence - Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene and 64% 1,4-butadiene incorporation, then used it in rubber compositions. - Example 2: Produced a similar copolymer with 25.7% styrene and 64% 1,4-butadiene incorporation, showing improved wear and modulus behavior in compositions. - Comparative formulation data: Demonstrated that replacing process oil with the tapered copolymer improved storage modulus and wear resistance. - Experimental section: Confirmed the copolymer is characterized by styrene sequence distribution, molecular weight, Tg, and viscoelastic performance.

Technical Relevance This is the most directly relevant patent for low styrene SBR because it explicitly claims 20 to 30 wt% styrene, with preferred ranges down to 22 to 27 wt%. It also emphasizes sequence distribution and low molecular weight, both important for tire-compound performance and oil replacement.

Patent 3

- Patent Number: US10676543B2
- Patent Title: Modified conjugated diene-based polymer, method for preparing the same and ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of a modified conjugated diene-based polymer by polymerizing styrene and 1,3-butadiene in n-hexane with an organometallic catalyst composition, then modifying the terminal active polymer with an alkoxysilane-based modifier. The examples show a styrene-butadiene random copolymer formed

at 90° C. and 50° C. with subsequent modification at 80° C., followed by isopropyl alcohol quench.

- Trans-1,4 bond content: 80 wt % or more — Claim 1 - Conjugated diene repeating unit: 50 to 95 wt % — Claim 2 - Aromatic vinyl repeating unit: 5 to 50 wt % — Claim 2 - Number average molecular weight: 10,000 to 500,000 g/mol — Claim 4 - Molecular weight distribution: 1.1 to 3 — Claim 4 - Styrene charge: 19.7 g — Example 1 - 1,3-Butadiene charge: 77.9 g — Example 1 - n-Hexane charge: 500 g — Example 1 - Barium di(ethylene glycol) ethyl ether: 1.3 mL of 30 wt % ethylbenzene solution — Example 1 - Trioctyl aluminum: 6.7 mL of 25 wt % hexane solution — Example 1 - n-Butyllithium: 1 mL of 2.5 M hexane solution — Example 1 - Reaction temperature: 90° C. — Example 1 - Polymerization conversion rate: 90% — Example 1 - Modifier amount: 0.4 g — Example 1 - Modification temperature: 80° C. — Example 1 - Modification time: 30 minutes — Example 1 - Styrene charge: 20 g — Comparative Example 2 - 1,3-Butadiene charge: 80 g — Comparative Example 2 - n-Hexane charge: 500 g — Comparative Example 2 - Tetramethylethylenediamine: 0.2 g — Comparative Example 2 - Reaction temperature: 50° C. — Comparative Example 2 - n-Butyllithium: 1 ml of 2.5 M hexane solution — Comparative Example 2 - Polymerization conversion rate: 98% — Comparative Example 2 - Modifier amount: 0.4 g — Comparative Example 2 - Modification temperature: 80° C. — Comparative Example 2 - Modification time: 10 minutes — Comparative Example 2 - DIN loss weight: 141 mg — Comparative Example table - DIN loss weight: 138 mg — Comparative Example table - DIN loss weight: 143 mg — Comparative Example table - DIN loss weight: 139 mg — Comparative Example table - DIN loss weight: 167 mg — Comparative Example table - DIN loss weight: 148 mg — Comparative Example table - DIN loss weight: 144 mg — Comparative Example table - Tan δ at 0° C.: 0.682 — Comparative Example table - Tan δ at 0° C.: 0.703 — Comparative Example table - Tan δ at 0° C.: 0.694 — Comparative Example table - Tan δ at 0° C.: 0.633 — Comparative Example table - Tan δ at 0° C.: 0.687 — Comparative Example table - Tan δ at 0° C.: 0.655 — Comparative Example table - Tan δ at 0° C.: 0.660 — Comparative Example table - Tan δ at 60° C.: 0.161 — Comparative Example table - Tan δ at 60° C.: 0.152 — Comparative Example table - Tan δ at 60° C.: 0.154 — Comparative Example table - Tan δ at 60° C.: 0.196 — Comparative Example table - Tan δ at 60° C.: 0.155 — Comparative Example table - Tan δ at 60° C.: 0.177 — Comparative Example table - Tan δ at 60° C.: 0.169 — Comparative Example table - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic hydrocarbon solution polymerization in n-hexane

styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Produced a styrene-butadiene random copolymer at 90° C. in n-hexane, then modified it with an alkoxysilane-based modifier at 80° C. for 30 minutes. - Example 2: Repeated the same process with a different alkoxysilane modifier, supporting the generality of the modification route. - Example 3: Added tetramethylethylenediamine as a polar additive to the same polymerization/modification sequence. - Comparative Example 2: Showed a lower-temperature polymerization at 50° C. and shorter modification time, used for comparison of trans-1,4 content and properties. - Comparative results:

Demonstrated that higher trans-1,4 content and silane modification improved abrasion, wet-road resistance, and rolling resistance balance.

Technical Relevance This patent is relevant because it directly discloses styrene-butadiene random copolymer synthesis and modification, with explicit styrene/butadiene feed amounts and a solution-polymerization route. While the target is not specifically low styrene SBR, the disclosed random copolymer embodiment and composition control are central to low styrene SBR development.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content	Key Findings
US10174133B2	Lg Chem, Ltd.	2019	Core polymerization is carried out in normal hexane solution. Water is used later only when the polymer is added to warm water heated using steam to remove solvents, followed by removal of remaining solvents and water by roll drying.	Not disclosed in extracted evidence	Example 1: Produced a modified styrene-butadiene copolymer from styrene/butadiene in hexane at 50° C. with n-butyllithiu

US11591428B2	Bridgestone Corporation	2023	Core polymerization is solution-based in a hydrocarbon solvent. Water is not the polymerization medium; it appears later only in coagulation workup through isopropanol-containing coagulation and drum drying of the recovered polymer.	20 to 30 percent by weight; preferred 22 to 27 percent by weight — Claim 1, Claim 2	Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene and 64% 1,4-butadiene incorporation, then u
US10676543B2	Lg Chem, Ltd.	2020	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1: Produced a styrene-butadiene random copolymer at 90° C. in n-hexane, then modified it with an alkoxysilane-ba

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US11591428B2 is the only patent that explicitly claims a low-styrene range for the target material, with total styrene content of 20 to 30 wt% and preferred 22 to 27 wt%.
- US10174133B2 and US10676543B2 both use solution polymerization of styrene and butadiene in hydrocarbon solvent, but neither explicitly discloses the styrene content of the target copolymer in the extracted evidence.
- US11591428B2 emphasizes tapered sequence distribution and low molecular weight (Mn 30,000 to 80,000), whereas US10174133B2 focuses on post-polymerization silane modification and US10676543B2 focuses on trans-1,4-rich conjugated diene structure plus alkoxy silane modification.
- All three patents rely on hydrocarbon-solvent solution processes rather than emulsion polymerization for the disclosed examples.
- Water is used only in downstream recovery/workup in US10174133B2; US11591428B2 uses coagulation in isopropanol, and US10676543B2 discloses alcohol quench without water in the extracted evidence.

4. CONCLUSION

Across the selected primary patents, the strongest direct low-styrene disclosure is US11591428B2, which claims a styrene-butadiene copolymer containing 20 to 30 wt% styrene and preferred 22 to 27 wt% styrene, with controlled styrene sequence distribution and low molecular weight. The other two patents are technically relevant because they disclose solution-polymerized styrene-butadiene copolymers and post-polymerization functionalization routes that are compatible with composition control, but they do not explicitly establish a low-styrene target in the extracted evidence. Collectively, the patents show that low-styrene SBR development in this set is centered on hydrocarbon-solution polymerization, controlled monomer feed/sequence architecture, and downstream functionalization or compounding to improve tire-relevant properties.

5. REFERENCES

- US10174133B2 | Modified styrene-butadiene copolymer, preparation method thereof, and rubber ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>
- US11591428B2 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Bridgestone Corporation | US | 2023 | <https://patents.google.com/patent/US11591428B2/en>
- US10676543B2 | Modified conjugated diene-based polymer, method for preparing the same and ... | Lg Chem, Ltd. | US | 2020 | <https://patents.google.com/patent/US10676543B2/en>